

Model Paper - Database Management Systems

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Q1 (20 marks)

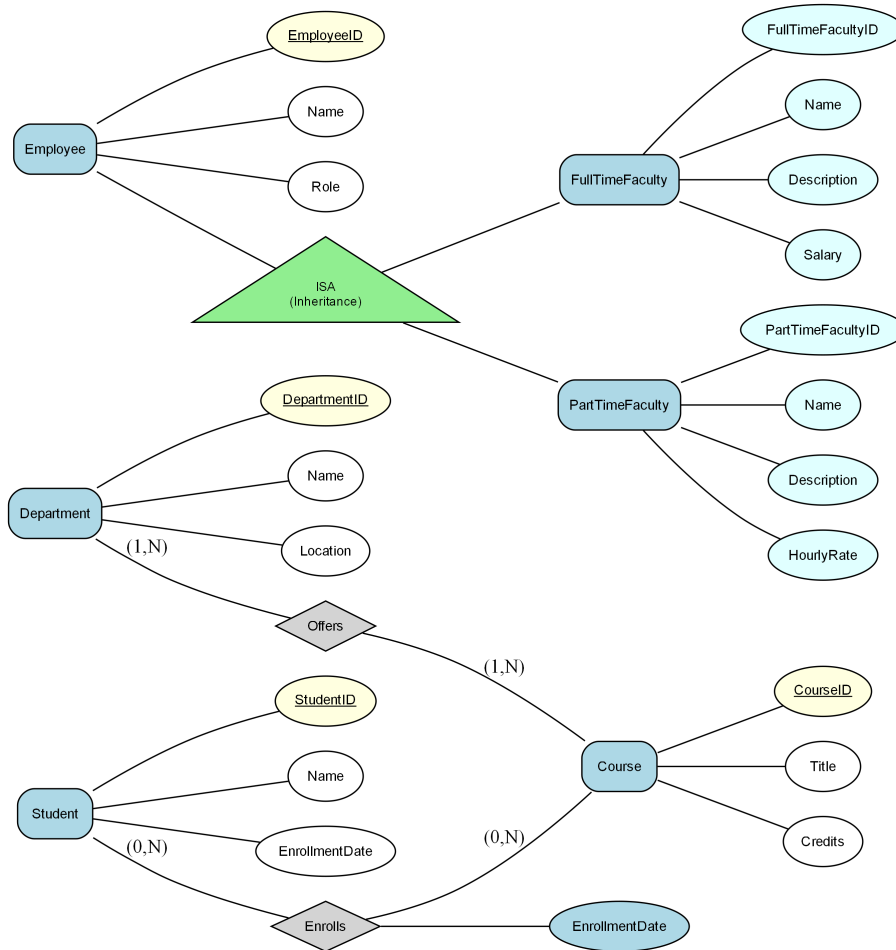
Consider the following Enhanced Entity-Relationship (EER) diagram:

Note: The diagram uses (min, max) cardinality notation where:

- (1,1) indicates one-to-one relationship (each entity participates exactly once)
- (1,N) or (1,*) indicates one-to-many relationship (one entity can relate to many)
- (0,1) indicates optional participation (zero or one)
- (0,N) or (0,*) indicates optional many participation (zero or many)

A university has several departments, each responsible for multiple courses. The 'Department' entity contains: DepartmentID, Name, and Location. Each 'Course' has attributes: CourseID, Title, and Credits. The university also has students enrolled in courses, represented by the 'Student' entity, which includes StudentID, Name, and EnrollmentDate. Moreover, the university employs teaching staff categorized under an 'Employee' entity with attributes: EmployeeID, Name, and Role. There are ISA hierarchies to distinguish between FullTimeFaculty and PartTimeFaculty under the Employee category. The FullTimeFaculty entity has attributes such as FullTimeFacultyID, Name, Description. includes Salary, whereas The PartTimeFaculty entity has attributes such as PartTimeFacultyID, Name, Description. includes HourlyRate. Each course can have multiple students enrolled, and students can enroll in multiple courses, establishing a many-to-many relationship with a descriptive attribute called EnrollmentDate, indicating when the student enrolled in the course. A Department can be related to many Courses through the 'Offers' relationship. A Student can be related to many Courses, and a Course can be related to many Students through the 'Enrolls' relationship (Student participation is optional, Course participation is optional)

Figure: EER



- Convert the following EER model into the relational model. Indicate the primary keys and the foreign keys of the resulting relations clearly. (17 marks)
- What is the best option for mapping the ISA hierarchies in the diagram? Justify your answer. (3 marks)

Q2 (15 marks)

Consider a relation $R(A, B, C, D, E)$ with the following set of functional dependencies F over R :
 $F = \{ \{AB \rightarrow C, C \rightarrow D, D \rightarrow E, E \rightarrow A, B \rightarrow AC\} \}$.

- Find all the keys in relation R using the attribute closure method. (4 marks)
- Is R in 3NF? Give reasons for your conclusion. (3 marks)
- Is R in BCNF? Give reasons for your conclusion. If R is not in BCNF, convert it to a set of BCNF relations. (8 marks)

Q3 (25 marks)

A retail company is developing a new database system to manage its inventory and sales data. The database will contain several tables, including Products, Categories, and Sales. Each Product has a unique identifier, name, price, and belongs to a Category. The Categories table contains information about product categories such as CategoryID and CategoryName. The Sales table records every sale transaction, which includes SaleID, ProductID, quantity sold, and sale date. The database is expected

to handle various operations including querying products based on price and category, as well as reporting total sales per category.

a) In Type 2 Driver, JDBC API calls are converted to native Java API calls. Accept or refute the above statement justifying your answer. (2 marks)

b) Code Segment: `java String sql = "SELECT * FROM employees WHERE department = ?";
PreparedStatement pstmt = connection.prepareStatement(sql); pstmt.setString(1, "IT"); ResultSet rs =
pstmt.executeQuery();` Which type of JDBC statement is used in the code segment shown above? Briefly explain when this type of statement will be used. (2 marks)

c) Briefly explain the options available for transferring data between two sources such as between tables and servers. (2 marks)

d) Briefly explain how referential integrity is maintained in a relational database management system. (3 marks)

e) An organization is establishing role-based access control for its database system. Sarah is the senior database administrator responsible for overall database administration. Emily is a junior database administrator responsible for managing the operational database. Nathan is a database developer responsible for creating and maintaining database objects. Michael is a data entry operator responsible for entering and updating transaction data.

i. Write a T-SQL statement to create a login to Sarah with windows authentication. (2 marks)

ii. Provide Sarah with the necessary permissions to handle all administrative tasks within the server using a fixed server role. (3 marks)

iii. Assuming Emily's username is 'emily.e', write T-SQL statements to assign her the responsibility of managing the operational database using a user-defined server role. (5 marks)

iv. Assuming Nathan's login name is 'nathan.n', write T-SQL statements to grant him permission to create tables and databases. (3 marks)

v. Assuming Michael's login name is 'michael.m', write T-SQL statements to allow him to perform data entry tasks on the operational databases. (3 marks)

Q4 (40 marks)

Consider the following schema of a database designed for an E-commerce platform: Product (productId: int, productName: varchar(100), price: real, stockAmount: int) Order (orderId: int, productId: int, quantity: int, orderDate: date) Customer (customerId: int, customerName: varchar(100), email: varchar(50), phone: varchar(15)) Payment (paymentId: int, orderId: int, paymentAmount: real, paymentStatus: varchar(20)). The 'Product' table stores information about products available for sale, including unique product ID, name, price, and stock available. The 'Order' table keeps track of customer orders, linking products with the quantity ordered and date of order. The 'Customer' table holds details of customers including their names and contact information. The 'Payment' table records payment details associated with each order, including payment amount and status. The 'Order' table holds order records, tracking customer orders with order ID, customer ID, order date, and total amount. The 'Payment' table contains relevant information for the database system.

a) Write SQL Queries to perform the following: i. Find the product name and price of products that have less than 10 items in stock.

- i. Find the product name and price of products that have less than 10 items in stock. (4 marks)
 - ii. Find the customer who has the highest total payment amount compared to all other customers. Find the customerId and customerName. (6 marks)
 - iii. Find the productName and customerName from Product, Customer, and Order where Order.orderDate is not null. (7 marks)
- b) b) Create a function to calculate the total payment amount for a given customer. This function should account for payment with a late penalty of 10% applied to those marked as 'overdue' payment status. (11 marks)
- c) c) Create a trigger that automatically updates the 'Payment' column in the 'Payment' table whenever a new payment record is added, or an existing payment record is updated. The 'Payment' column should contain the total payment amount for each customer. Ensure the trigger adjusts the 'Payment' column accurately to reflect any partial payments made by customer. The customer pays only 50% of the total payment amount as an initial payment when the payment status is marked as 'Partial'. (12 marks)